

# Linear Regression Report

Task 1 — House Price Prediction · California Housing Dataset  
Maincrafts Technology | AI & ML Internship

| Dataset                           | Features | Train Samples | Test Samples | Algorithm         |
|-----------------------------------|----------|---------------|--------------|-------------------|
| California Housing (scikit-learn) | 8        | 16,512        | 4,128        | Linear Regression |

## 1. Objective

The goal of Task 1 was to go through a complete, end-to-end Machine Learning workflow — from raw data loading through exploratory analysis, model training, evaluation, and saving. A Linear Regression model was trained to predict median California house prices (in units of \$100k) from 8 input features.

## 2. Exploratory Data Analysis (EDA)

### 2.1 Dataset Overview

The California Housing Dataset contains 20,640 rows and 9 columns (8 features + 1 target). There are no missing values, which made preprocessing straightforward. The target variable **MedHouseVal** represents median house value and is capped at \$500k — visible as a spike in the distribution at 5.0.

### 2.2 Feature Summary

| Feature    | Description                | Notable Observation                               |
|------------|----------------------------|---------------------------------------------------|
| MedInc     | Median block income        | Strongest correlation with target (0.688)         |
| HouseAge   | Median house age (years)   | Fairly uniform distribution, range 1–52           |
| AveRooms   | Avg rooms per household    | Right-skewed; outliers up to 141 (data issue)     |
| AveBedrms  | Avg bedrooms per household | Correlated with AveRooms — multicollinearity risk |
| Population | Block population           | Heavily right-skewed; max ~35,682                 |
| AveOccup   | Avg household occupancy    | Outliers > 10 present (~500 rows)                 |
| Latitude   | Geographic latitude        | Negative correlation with price (-0.144)          |
| Longitude  | Geographic longitude       | Negative correlation with price (-0.046)          |

### 2.3 Key Findings from EDA

- MedInc** is by far the most predictive feature ( $r = 0.688$ ). Wealthier neighbourhoods consistently have higher house values.

- Most features are **right-skewed** — Population and AveOccup especially. This hints that a linear model may struggle.
  - There is a **hard ceiling at \$500k** in the target. This introduces artificial patterns and likely hurts model performance at the upper end.
  - **AveRooms and AveBedrms are highly correlated** with each other (~0.85), introducing mild multicollinearity.
  - Latitude shows a clear geographic signal — southern California (lower latitude) tends to command higher prices.
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## 3. Model — Linear Regression

### 3.1 Approach

Ordinary Least Squares Linear Regression was used as the algorithm. All 8 features were included without feature selection or scaling (scaling is addressed in Task 2). The dataset was split 80/20 into training and test sets using *random\_state=42* for reproducibility.

### 3.2 Feature Coefficients

After training, the model assigned the following coefficients. Note: because features are not scaled, raw coefficient magnitude does not directly reflect importance — a large coefficient on Latitude simply reflects its small numeric range.

| Feature    | Coefficient | Direction                               |
|------------|-------------|-----------------------------------------|
| MedInc     | 0.4490      | Positive — higher income → higher price |
| AveRooms   | 0.1100      | Positive                                |
| Latitude   | -0.8975     | Negative — further north → lower price  |
| Longitude  | -0.8680     | Negative                                |
| HouseAge   | 0.0118      | Positive — older homes slightly pricier |
| AveBedrms  | -0.1165     | Negative                                |
| Population | -0.0000     | Negligible                              |
| AveOccup   | -0.0036     | Negative                                |

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## 4. Model Evaluation Metrics

The trained model was evaluated on the held-out 20% test set using three standard regression metrics:

| Metric                     | Value  | Interpretation                      |
|----------------------------|--------|-------------------------------------|
| MAE (Mean Absolute Error)  | ~0.531 | Avg prediction error of ~\$53,100   |
| RMSE (Root Mean Sq. Error) | ~0.745 | Penalises large errors more heavily |

| Metric                                     | Value         | Interpretation                               |
|--------------------------------------------|---------------|----------------------------------------------|
| <b>R<sup>2</sup> (Coefficient of Det.)</b> | <b>~0.596</b> | <b>Model explains ~60% of price variance</b> |

#### 4.1 What These Numbers Mean

- An  $R^2$  of 0.596 is a solid baseline for a raw, unscaled dataset. It means the model captures the broad trend but misses finer non-linear patterns.
- The MAE of ~\$53k is meaningful — for a \$300k house that's a ~18% error. Acceptable for a first model, but not production-ready.
- RMSE being noticeably higher than MAE indicates the model makes some large errors — likely on the capped \$500k properties.

#### 4.2 Visual Analysis

The **Actual vs Predicted** scatter plot shows a positive linear trend broadly following the perfect-fit diagonal, but with increasing spread at higher values. The **Residual Plot** reveals a non-random pattern — residuals fan out as predicted value increases (heteroscedasticity), and there is a visible flat band near predicted=5.0 caused by the \$500k cap in the target variable. Both patterns confirm the model is hitting the limits of what a linear approach can achieve here.

### 5. Improvement Ideas

The following enhancements were identified during the analysis. Each addresses a specific limitation observed in the EDA or evaluation:

| Improvement                                | Why It Helps                                                                | Expected Gain                                 |
|--------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------|
| Feature Scaling (StandardScaler)           | Puts all features on equal footing; required for Ridge/Lasso regularisation | Modest on R2; critical for regularised models |
| Outlier Capping (AveRooms, AveOccup)       | Extreme values (~141 rooms) skew the model and inflate RMSE                 | Modest RMSE reduction                         |
| Polynomial Features                        | Captures non-linear relationships (e.g. income vs price is curved)          | +5–10% R2                                     |
| Ridge / Lasso Regression                   | Reduces overfitting; Lasso also performs automatic feature selection        | Marginal improvement on this dataset          |
| Random Forest or XGBoost                   | Tree ensembles handle non-linearity and interactions naturally              | R2 can reach 0.80–0.85                        |
| k-Fold Cross-Validation                    | Single 80/20 split can be lucky; k-fold gives a more reliable estimate      | More trustworthy metrics                      |
| Feature Engineering (income x rooms ratio) | Domain-driven features can expose relationships the raw features miss       | Moderate R2 uplift                            |

### 6. Conclusion

The Linear Regression model provides a functioning baseline with  $R^2 \sim 0.60$  and MAE  $\sim \$53k$ . It correctly identifies **median income** as the dominant driver of house prices and captures the broad geographic trend via latitude and longitude. However, residual analysis clearly shows the model is limited by non-linearity and the \$500k cap in the target.

Task 2 addresses these limitations directly through feature scaling and multi-model comparison. The most significant performance jump is expected from moving to an ensemble method (Random Forest / XGBoost) in a future task.